



Influence by osmosis: Social media green communities and pro-environmental behavior

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ABSTRACT

While there is growing adoption of social media by environmental stakeholders, whether and how social media green communities affect netizens' pro-environmental behavior is understudied. Underpinned by social learning theory, this study explores the relation of green community browsing to netizens' pro-environmental behavior, the underlying mechanisms, and the interactive role of a critical social cue. Results of structural equation modeling in a sample of 585 members of green communities indicate that green communities' browsing has a significant relationship with pro-environmental behavior through the sequential mediation of participation and environmental concern. The volume of likes strengthens the relationship between netizens' browsing of green communities and participation. This novel study offers valuable insights into the role of social media green communities.

1. Introduction

Social media users are mounting worldwide as 2021 witnessed 424 million new users, with a 10% growth (Gaboriault, 2022). Social media browsing is a leading online activity for 4.6 billion netizens, constituting 58% of the entire population, predominantly manipulating their lives (Gaboriault, 2022). Various communities have been established on social networking platforms (e.g., brand communities, traveling communities, influencer communities, and ethical communities) that are trying to influence netizens' specific behaviors while browsing, such as brand awareness, choice of travel destination, informed shopping decisions, and ethical consumption (Gummerus, Liljander, & Sihlman, 2017; Hajli, 2018; Kamboj & Rahman, 2017; Nadeem, Juntunen, Hajli, & Tajvidi, 2019). One influential community among netizens is a green community created by environmental stakeholders (Zafar, Shen, Shahzad, & Islam, 2021). It can be defined as "affiliative groups whose online interactions are based upon a shared enthusiasm for, and knowledge of, a specific consumption activity or related group of activities" (Kozinets, 1999, p. 254). According to Zafar, Shen, Ashfaq, and Shahzad (2021), green

communities on social media have been formed to promote pro-environmental behavior, which is "any action that enhances the quality of the environment, either resulting in or not resulting from pro-environmental intent" (Steg, Bolderdijk, Keizer, & Perlaviciute, 2014, p. 29).

A growing literature investigates the impact of media (i.e., television, newspapers, and the internet) on users' pro-environmental behaviors, and it is suggested that every media has unique features, importance, and level of involvement (Huang, 2016). Social media can be more impactful than traditional means due to the abundant information and high levels of involvement and interactions (Chung, Chiu, Ho, & Au, 2020; Liu, Han, & Teng, 2021). Scholars, for instance, Chen, Ghosh, Liu, and Zhao (2019) highlighted the lack of research incorporating contemporary media's role in behaviors, which is vital for society and can assist in achieving sustainable development. Further, the role of particular media and content needs to be investigated for theoretical development, and administrative implications rather than general ones in the context of pro-environmental behavior since individuals consume specific content on distinctive platforms (Liu et al., 2021). Han, McCabe,

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Wang, and Chong (2018) and Hynes and Wilson (2016) have examined the role of social media in pro-environmental behavior through personal and social norms and eco-friendly user-generated content. The existing literature presents a superficial understanding of how social media influences netizens' pro-environmental behavior without scientific elucidation and the underlying mechanisms. These findings cannot be generalized in the context of green communities since such platforms have their own importance, and every user is not part of it (Zafar, Qiu, Li, Wang, & Shahzad, 2019). Researchers have shed light on green community initiatives in the real world, which people prefer, and they participate in different environmental activities (Boland & Zhu, 2012; Hostetler, Swiman, Prizzia, & Noiseux, 2008). Consumers like to become part of green communities, influencing their ecological knowledge and shaping their behaviors (Cegarra-Navarro, Cordoba-Pachon, & de Bobadilla, 2009; Hostetler et al., 2008). There are few studies conducted on social media and offline green communities; however, research incorporating the role of social media green communities in users' pro-environmental behavior is still in its infancy. The current study investigates the ascendancy of green communities in members' pro-environmental behavior and the underlying mechanisms.

Our study is underpinned by social learning theory, which posits that people's deeds are influenced by and can be learned from observing and imitating the behavior of others (Bandura and McClelland, 1977). Social media communities are the modern paradigm of social learning in which the learner interacts with the material, social or cultural environment, leading to intrinsic processes of procurance and amplification (Gummerus et al., 2017; Hajli, 2018; Lorenzo, Kawalek, & Ramdani, 2012). The present research argues that green communities promote pro-environmental behavior because browsing these communities helps them to access better ecological information and learn from other experiences. Browsing will attract netizens to participate in the communities and increase environmental concerns, referring to "an affective attribute that can represent a person's worries, compassion, likes and dislikes about the environment" (Lee, 2009, p. 88). Green communities are associated with pro-environmental behavior through these underlying mechanisms and other interactive features, such as the volume of likes.

This study offers novel insights into the literature through the proposed framework following the social learning theory. First, the importance of browsing green communities is stated in endorsing pro-environmental behavior through a systematic evaluation of contextual determining factors. Second, participation and environmental concerns are integrated into the underlying mechanism, and their simple and sequential mediation is also unveiled for better understanding. Third, the critical numeric cue of social media (i.e., the volume of likes) may strengthen participation's associations with environmental concerns due to its circumstantial importance and provide valuable insights into the boundary conditions moderating the indirect relationships between browsing green communities and pro-environmental behavior. In summary, this research provides a comprehensive insight into the social and psychological processes through which green communities exert influence on netizens' pro-environmental behavior. Approaching sections clarify the theoretical foundation, hypotheses, methodology, findings, implications, and suggestions for future researchers, along with limitations.

2. Theoretical foundation and hypotheses development

2.1. Social learning theory and green communities on social media

Social media enables the users to become part of embedded communities, having distinctive importance within these podiums (Gummerus et al., 2017; Zafar, Qiu, & Shahzad, 2020). Different communities co-create, alter and disseminate info that influences different kinds of citizenship behavior of public and corporates (Whelan, Moon, & Grant, 2013). These communities assist in developing interactive relationships

between netizens, organizations, and each other (Gummerus et al., 2017). Usage (Hyun, Thavisay, & Lee, 2021; Leong, Jaafar, & Ainin, 2018) and browsing (Weinstein, 2017; Zhang, Xu, Zhao, & Yu, 2018) are two distinctive elements of social media. Browsing elucidates netizens' activities rooted in exploratory behavior, wherein they encounter posts with strong and weak ties, including communities (Weinstein, 2017; Zhang et al., 2018), and usage reveals the amount of time a user spends on social media (Hyun et al., 2021; Leong et al., 2018). The current study focuses on browsing in the context of green communities considering its scope and relevancy. A few illustrations of green communities' posts are provided in the Appendix.

Users may express diverse views on social media; however, we employ the optimistic approach from the social learning perspective to explore the role of green communities in promoting pro-environmental behavior. Netizens learn from social media communities, assessments, and endorsements that affect their psychological processes, such as observation, imitation, and decision-making (Chen, Lu, & Wang, 2017; Hajli, 2018). Imitation designates the effectual modeling procedure as persons replicate analogous conduct (Lorenzo et al., 2012; O'Fallon & Butterfield, 2012). This illustrates the learning rooted in the social learning theory suggested by Bandura and McClelland (1977). It provides a structured framework for dealing with different behavioral concerns and human development (Chen et al., 2017). Online platforms can support pro-environmental actions as environmental information embedded in user-generated social media content can enhance environmental norms and behavioral intents (Han et al., 2018). For instance, netizens can support different pro-environmental campaigns by liking or sharing these activities, disseminating such acts among their social network, and promoting online donation activities (Lee et al., 2020). Realizing the importance of contemporary media, many celebrities portray themselves as eco-friendly and use social media to present eco-friendly issues to a large community (Hello, 2017; Smith, 2013).

Pro-environmental behavior comprises numerous characteristics that make it suitable to support the development of green communities. Pro-environmental behavior is a cognitively driven act and has a social domain because it links individuals with society (Alzubaidi, Slade, & Dwivedi, 2020). These associations may be on both the micro- and -macro levels, involving families, markets, and states (Alzubaidi et al., 2020). Netizens with environmental concerns consider themselves liable to society and communicate their thoughts on individuals' acts (Ashfaq, Zhang, Ali, Waheed, & Nawaz, 2021). Green communities encourage netizens to share their eco-friendly choices and ideologies with others; thus, established platforms can be vital elements in promoting pro-environmental behavior. Therefore, it can be suggested that in line with social learning theory, green communities are crucial platforms for promoting environmental behavior.

2.2. Impact of browsing green communities

Online browsing is an explorative "search behavior characterized by the user actively scanning an environment when moving through it" (Pöyry, Parvinen, & Malmivaara, 2013, p. 227), and social media browsing scans the newsfeed for information or entertainment purposes (Zhang et al., 2018). The browsing of online communities instigates individuals to become involved in interactive activities with other users (Lu, Zhao, & Wang, 2010). In a social media environment, netizens learn from their peers' knowledge and experiences (Chung et al., 2020). Ratings and recommendations of others in online communities influence users' cognition and behavior (Chen et al., 2017; Hajli, 2018). While browsing, followers of social media communities are inclined to create content in that environment rather than remain passive (Shanmugam, Sun, Amidi, Khani, & Khani, 2016; Tajvidi, Richard, Wang, & Hajli, 2018) to facilitate information exchange and social value-sharing (Bateman, Gray, & Butler, 2011). Generating content within a community is regarded as member participation and is entrenched in the browsing experience. Browsing promotes such activities, and social

media users are motivated to contribute and participate actively, which is weighed as a tangible symbol of devotion (Casaló, Flavián, & Guina-líu, 2010; Pöyry et al., 2013). Members of online communities tend to form social groups to support each other, and these groups become stronger when members' participation increases (Zafar et al., 2020). A similar perspective can be implemented in the context of green communities.

Valkenburg (2021) and Weinstein (2017) have highlighted the importance of social media browsing in netizens' psychological well-being based on perceptive factors and environmental actions while perusing others' posts. Browsing may attract users to do online environmental activities and create content in green communities; as such kind of virtual communities have been encouraging netizens to browse and participate in attaining their objectives associated with ethical actions, online buying, electronic word of mouth, and digital celebrities (Hajli, 2018; Pöyry et al., 2013; Ruiz-Mafe, Bigne-Alcañiz, Sanz-Blas, & Tronch, 2018; Zafar et al., 2020). Communities tend to persuade their members to disseminate opinions and knowledge with peers, which shapes their cognitive and affective strategies (Zhang & Watts, 2008). Similarly, many individuals who follow different virtual communities engage with encountered posts while browsing, associated with their participative activities (Hajli, 2018; Zafar et al., 2020). Particularly, the response of peers on a post not only encourages the similar person to carry on but it also encourages others to participate; accordingly, they may feel respected, valued, and satisfied in a virtual community though the response can be positive or negative; however, it assists in building a social climate, meant to interaction and dedicated support to do any act (Zafar et al., 2020). It can be argued based on the above literature that participation originating from browsing can assist the members in identifying with green communities and becoming interested in participating in online interactions with other community members; therefore, current research suggests the following hypothesis:

H1a. Browsing green communities is positively related to members' participation in these communities.

Users browse embedded online communities, which persuade their followers to share opinions and knowledge with their peers, shaping their perceptions and behaviors regarding different activities (Gummerus et al., 2017; Zafar et al., 2020). The information generated by peers and encountered while browsing may be more influential than that generated by organizations (Füller, Bartl, Ernst, & Mühlbacher, 2006). Environmental information embedded in user-generated content on social media can significantly enhance environmental concerns and behavioral intentions in the tourism sector (Han et al., 2018). The literature reflects the importance of social determinants in the virtual environment (Zafar, Shen, Ashfaq, & Shahzad, 2021). Different embedded virtual social cues impact environmental intentions and eco-friendly decision-making because individuals receive social support and feel pressure from those they have strong ties, such as family and close friends (Hartmann, Fernández, Apaolaza, Eisend, & D'Souza, 2021; Johnstone & Hooper, 2016; Ridings, Wasko, Oinas-Kukkonen, Lyytinen, & Yoo, 2010).

The virtual environment also offers e-social norms (de Kervenoael, Aykac, & Palmer, 2009). This normative motivation and pressure may be more potent for netizens, particularly among the followers of green communities. Modern technologies such as social media, web, and mobile technologies motivate people to care about the environment and increase environmental awareness (Han et al., 2018; Wheaton et al., 2016). Further, Lee et al. (2020) posited that different environmental cues embedded in social media platforms might enhance users' attitudes toward recycling, rooted in environmental concerns. In a similar vein, it can be argued that browsing green communities will enhance ecological concerns, as Zhang et al. (2018) postulated that many social media users perform utilitarian browsing to influence their future decisions. Green communities contain ample user-generated content and strive to promote environmental concerns by publishing pro-environment posts that

users encounter as they browse. Therefore, this study proposes the following hypothesis:

H1b. Browsing green communities is positively related to members' environmental concerns.

2.3. Role of participation in green communities

Participation in online communities involves content creation, highlighting their interactive and contributive usage behavior (Casaló et al., 2010). In virtual communities, participation is the foundation of building, attaining, and sharing knowledge to create and achieve values (Gummerus et al., 2017; Wiertz & de Ruyter, 2007). Virtual community participation has "actionable," "informational," and "attitudinal" dimensions (Kamboj & Rahman, 2017). According to Kamboj and Rahman (2017) and Nadeem et al. (2019), actionable participation represents the level at which members regularly contribute to activities and the level of interactions between users. Informational participation deals with obtaining information and satisfying community members' general interest in a service or product. Finally, attitudinal participation represents members' propensity to assess a community's performance positively or negatively and includes optimistic or adverse attitudes toward the platform, product, or service.

Users support environmental activities through several forms of participation (Lee et al., 2020). Social media may be a significant driver in enhancing information regarding the environment (Chung et al., 2020). Participation in green communities may provide access to more insightful information about environmental issues and protection. Accordingly, members comprehend the importance of ecological fortification better and are more likely to accept pro-environment views. Nadeem et al. (2019) have accentuated the prominence of participation in sharing economy platforms and suggested that these may influence value co-creation and participants' ethical perceptions. Similarly, contribution to green communities will augment participants' environmental concerns as online interpersonal social influence positively stimulates users' green shopping decisions (Bedard, Tolmie, Anne, Bedard, & Tolmie, 2018). Building on the above discussion, we postulate that participation in green communities will promote environmental concerns. Therefore, we develop the following hypothesis:

H2. Participation in green communities is positively related to environmental concerns.

2.4. Mediating effect of participation in green communities and environmental concerns

Environmental concern is considered a fundamental predictor of decision-making in ecological research (Felix, Hinsch, Rauschnabel, & Schlegelmilch, 2018; Lee, Kim, Kim, & Choi, 2014), and environmentally friendly behavior is ingrained in individuals' environmental concerns (Cerri, Testa, & Rizzi, 2018; Hynes & Wilson, 2016). It is argued that environmental concerns lead to pro-environmental behavior, including green purchases and environmentalist behavior, both directly and indirectly via different social and psychological routes (e.g., Alzubaidi et al., 2020; Polonsky, Kilbourne, & Vocino, 2014; Yadav & Pathak, 2016). In brief, environmental concern has a strong potential to influence several domains of ecological behavior, such as donations for environmental protection, sustainable shopping, support of environmental activists, and the adoption of a low-carbon lifestyle. The connection between ecological concerns and pro-environmental actions has been well studied. Hypotheses 1a, 1b, and 2 suggest positive associations between browsing green communities, participation in green communities, and environmental concern. Taken together, environmental concern is a mediator through which browsing green communities and participation in these communities are linked to pro-environmental behavior. Therefore, we hypothesize the following:

H3a. Environmental concern mediates the association between browsing green communities and pro-environmental behavior.

H3b. Green community participation and environmental concern sequentially mediate the association between browsing green communities and pro-environmental behavior.

2.5. Moderating role of the volume of likes

The volume of likes illustrates the number of positive rejoinders a post receives and is considered one of the critical features of social networking platforms (Zafar, Qiu, et al., 2021). Such observational learning enhances awareness among potential consumers, as they know about the number of informed customers in a market (Liu, 2006). Ye, Law, Gu, and Chen (2011) suggested that the users' activities (e.g., wish list, volume of sales, espousal ratio, "share" and "like" buttons) will inform a new shopper regarding other purchasers and their activities. Embracing a similar perspective, it was posited that the volume of likes significantly impacts social media users' wishes (Chen, Su, & Widjaja, 2016). Knowledge attained through observational learning is perceived as credible, as "actions usually speak louder than words" (Chen et al., 2016). Furthermore, people do not have unrestrained intellectual capacity and often adopt a simplified and heuristic approach to obtaining information and making decisions (Hu, Koh, & Reddy, 2014). The volume of likes may provide discrete gestures to peers, as articulated by observational learning. While browsing green communities, social media users may encounter posts that may have a higher volume of likes than other posts. A higher volume of likes will be more influential and persuasive. People have tight schedules in their daily lives and attention-seeking works; therefore, are more likely to engage with posts that have more likes as they browse. Therefore, the relationship between browsing green communities and community participation is likely to be contingent on the volume of likes. Thus, this study proposes the following hypothesis:

H4. The volume of likes moderates the relationship between browsing green communities and community participation such that the higher (lower) volume of likes of the encountered posts, the stronger (weaker) the relationship.

Collectively, these hypotheses indicate that browsing green communities indirectly influences pro-environmental behavior, as if by osmosis, through the mediating effect of participation in these communities and environmental concerns. Moreover, the volume of likes moderates the impact of browsing green communities on participation in these communities. Fig. 1 portrays the research framework and hypotheses.

3. Research methodology

3.1. Measures

This study used existing instruments that have been empirically validated in previous studies. Integrated measures were anchored on a "seven-point Likert scale" ranging from 1 = "strongly disagree" to 7 = "strongly agree." A pre-test was performed with 50 followers of green communities before the formal survey was conducted. The survey was administered in English.

3.1.1. Browsing green communities

The browsing of green communities was measured using three items from Huang (2016). These items include "I like to browse the green communities for ideas," "The percentage of my time I spend looking around on green communities is fairly high," and "I like to browse the community to see what is new (either directly on the community page or through my newsfeed)." This browsing measure has good internal consistency in the extant literature ($\alpha = 0.90$; Leong et al., 2018). The internal reliability in the current research was 0.85.

3.1.2. Participation in green communities

Participation was assessed using the six items from Nadeem et al. (2019). Participation is a multidimensional construct with informational, actionable, and attitudinal dimensions. The informational dimension has two items: "I post messages and provide responses in green communities frequently" and "I frequently provide useful information in green communities to the other members." The actionable dimension was assessed with two items: "I actively participate in green community activities" and "I spend a lot of time participating in green community activities." The attitudinal dimension also had two items: "I think participating in green communities would be good for me" and "I think participating in green communities would be beneficial for me." Nadeem et al. (2019) demonstrated good internal consistency for the participation scale ($\alpha = 0.86$, informational; $\alpha = 0.93$, actionable; and $\alpha = 0.91$, attitudinal). The internal reliability values for informational, actionable, and attitudinal participation in this study were 0.82, 0.75, and 0.71, respectively.

3.1.3. Environmental concern

Environmental concern was assessed using four items following the recommendation of Lee et al. (2014). Sample items are "I am concerned about the condition of the environment" and "I am worried that the condition of the natural environment is getting worse every year." Lee et al. (2014) reported good internal reliability for this scale ($\alpha = 0.79$). The α value for this scale in this study was 0.82.

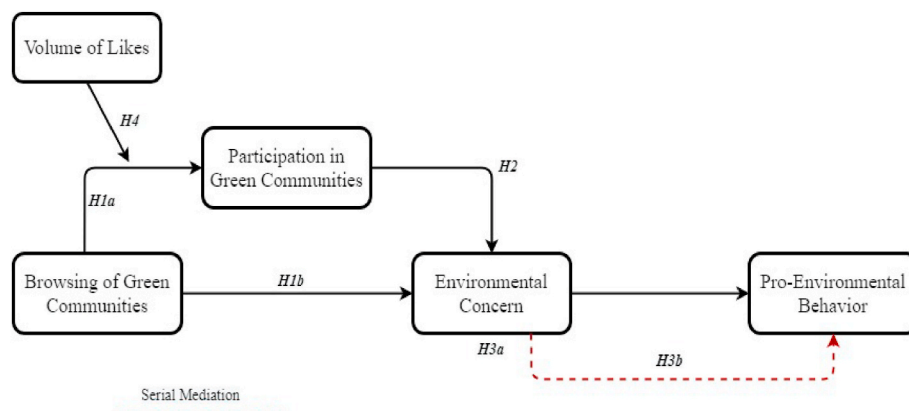


Fig. 1. The research model and hypotheses.

3.1.4. Pro-environmental behavior

Pro-environmental behavior was measured using seven items from Huang (2016). Sample items include “I turn off or unplug electronic devices when not needed,” “I reduce driving, and walk, cycle, or use public transportation,” and “I reduce using plastic bags or use my own bag when shopping.” Huang (2016) reported good internal consistency for the pro-environmental behavior scale ($\alpha = 0.91$). The internal reliability for pro-environmental behavior in this study was 0.85.

3.1.5. Volume of likes

The volume of likes was assessed using three items taken from Zafar, Qiu, et al. (2021). Sample items are “It is easy for me to observe that environmental posts are liked by many people” and “I observe that the quantity of likes for different posts is great.” Zafar, Qiu, et al. (2021) reported good consistency for the employed scale ($\alpha = 0.82$). In the existing research, the α for this scale was 0.88.

3.2. Control variables

Respondents' followed number of communities and usage of social media was reported to be linked to users' pro-environmental behavior (Kadic-maglajlic, Arslanagic-kalajdzic, Micevski, Dlacic, & Zabkar, 2019; Zafar, Shen, Shahzad, & Islam, 2021). These two variables were controlled for in the current study. Users' demographic attributes, such as gender, qualifications, occupation, and income, can affect pro-environmental behavior (Kadic-maglajlic et al., 2019). We, therefore, also controlled for these demographic attributes in the current study.

3.3. Data collection

Researchers have emphasized the importance of Pakistani social media communities in several fields (Zafar et al., 2020). It has been reported that 43.55 million individuals use Facebook, 13.75 million use Instagram, 18.26 million use TikTok, 3.40 million use Twitter, 18.80 million use Snapchat, and 7.60 million use LinkedIn in Pakistan (Kemp, 2022). Furthermore, the Pakistani government has focused on sustainable development, and several steps have been taken (e.g., the “Billion Tree Tsunami” project), which is recognized as a green revolution (Government of Pakistan, 2019; Mangi, 2020). Accordingly, environmental issues and different pro-environmental activities are widely shared on social media platforms by concerned stakeholders (The News, 2020). The population investigated in the current study was Pakistani social media users who follow green communities. The literature and facts outlined above demonstrate the value of exploring Pakistani followers of green communities in our proposed model.

The sample consists exclusively of green community members. To do so, a screening question was incorporated at the beginning of the questionnaire (i.e., do you consider yourself an active member of a green community on any social media platform?). Respondents who gave a “No” answer were not encouraged to complete the survey or be included in the late analysis. In the process of data collection, a two-step approach was performed wherein, initially, data were directly collected from followers of different green communities; accordingly, the link was shared with an essential condition imposed on each question to avoid missing values. Diverse time slots were chosen during the data collection period to ensure that all followers had the opportunity to participate in the survey. In the second stage, targeted respondents from green communities were approached utilizing social media features, which provided details regarding how many of each author's friends are following these communities. Further, these respondents were asked to circulate the questionnaire to generate a snowball effect and increase the sample size. The final sample consists of 585 respondents, and their demographic details are presented in Table 1. Two criteria assessed the sample size. First, the ten-times rule was followed, as Hair, Hult, Ringle, and Sarstedt (2017) recommended. Second, G*Power software was

employed to determine the adequacy of the sample size using power analyses (Prajapati, Dunne, & Armstrong, 2010).

4. Findings

4.1. Analytical approach

The data was analyzed by Smart PLS and SPSS software tools. Partial least squares structural equation modeling (PLS-SEM) has the potential to assess measurement and structural frameworks proficiently. A multifaceted framework with several variables and incremental characteristics (e.g., volume of likes—moderator; participation, and environmental concern—mediator) can be assessed without construct removal and data distribution assumptions (Hair et al., 2017). Moreover, PLS-SEM handles second-order formative constructs better and therefore is a recommended approach for forecasting and exploring endogenous variables to explain the maximum variance, especially when the sample is small (Hair et al., 2017).

4.2. Common method variance

The single-source data may result in common method variance (CMV) (Podsakoff, MacKenzie, & Podsakoff, 2012). Multiple methods were integrated to evaluate CMV. First, the results of Harman's single factor reveal the first-factor value, 30.59, is lower than 40% (Babin, Griffin, & Hair, 2016; Harman, 1976). Second, the marker approach was adopted to detect CMV (Lindell & Whitney, 2001; Malhotra, Kim, & Patil, 2006). CMV must originate the correlation between the marker and other variables. The value of shared correlation was derived from the average correlation of the targeted and the marker variable. In this manner, adjusted path coefficients and correlation were estimated, and no significant difference was seen relative to the originals. Third, collinearity was assessed to detect CMV, following Kock (2015). Values of the variance inflation factors (VIF) were lower from the threshold of 3.3. Lastly, social desirability bias (SDB) was determined. Seven items anchored on a Likert scale were incorporated in the survey to measure SDB, following previous studies (Larson, 2019; Zafar et al., 2020). The resulting values were low and inconsequential, demonstrating the absence of SDB. These findings show that serious CMV issues were not present in this study.

4.3. Measurement model

The descriptive statistics (e.g., standard deviations, means) of the targeted latent are stated in Table 2. The appropriateness of the measurements was assessed by considering reliability, discriminant validity (DV), and convergent validity (Hair et al., 2017). All items had factor loadings from 0.69 to 0.93 (given in Table 2), and all reflective constructs had average variance extracted (AVE) values not lower than 0.50. Further, composite reliability and Cronbach's alpha had values not lower than 0.70 (Cohen, 1988). These results imply good reliability and convergent validity for the reflective constructs.

In the next step, we evaluated the DV against the heterotrait-monotrait ratio (HTMT) (Henseler, Ringle, & Sarstedt, 2014) and the Fornell-Larcker criterion (Fornell & Larcker, 1981). The HTMT values were not higher than 0.85, and the highest value was 0.71, indicating the distinctiveness of all variables. According to Hair et al. (2017), HTMT should not have values higher than 0.90, while Henseler et al., 2014 have suggested a more cautious criterion of 0.85. The result of the Fornell-Larcker test is presented in Table 3, wherein highlighted values of AVE square root (diagonal ones) are higher than the associated values. Thus, the overall findings establish satisfactory results for DV.

Participation in green communities was integrated as a second-order formative construct. The existing body of knowledge argues that traditional methods are not appropriate for evaluating validity and reliability (Diamantopoulos & Winklhofer, 2001). This research gauged the outer

weights of first-order constructs as recommended by [Petter and Rai \(2007\)](#). Resultant values imply that all constructs had significant outer weights: actionable participation, $\beta = 0.40$, T statistics = 23.48; attitudinal participation, $\beta = 0.40$, T statistics = 29.94; and informational participation, $\beta = 0.44$, T statistics = 24.33. Moreover, the VIF values of all dimensions were less than 3.3, suggesting that participation in green communities is an effective formative construct. Collinearity issues may influence structural models; therefore, analyzing the VIF values of integrated variables was necessary. The VIF values were below the recommended threshold of 5.0 for targeted constructs and items ([Hair et al., 2017](#)). The correlation values of the principal variables are provided in [Table 4](#) and show that multicollinearity is not an issue. Overall, the results demonstrated that the measurement model was satisfactory and that inner model analysis could be undertaken.

4.4. Hypothesis testing

Structural model analysis and bootstrapping with 5000 resamples were employed to assess all hypotheses. All control variables were insignificant for pro-environmental behavior (e.g., gender ($\beta = -0.03$, $p = 0.460$), qualification ($\beta = 0.02$, $p = 0.621$), occupation ($\beta = -0.03$, $p = 0.384$) and income ($\beta = 0.04$, $p = 0.269$) as their p -values > 0.05 . Hypotheses 1a and 1b predicted that browsing green communities is positively related to participation in green communities and environmental concerns. The results revealed that browsing green communities has a noteworthy positive association with participation ($\beta = 0.43$, $p = 0.000$) and environmental concerns ($\beta = 0.24$, $p = 0.000$). Hence, [H1a](#) and [H1b](#) were supported. The result also supported Hypothesis 2, which predicted that participation in green communities directly correlates with environmental concerns ($\beta = 0.18$, $p = 0.000$).

To test Hypotheses 3a and 3b suggesting simple mediation and sequential mediation, we first analyzed the link between environmental concern and pro-environmental behavior. The results showed that environmental concern is positively related to pro-environmental behavior ($\beta = 0.41$, $p = 0.000$). Next, indirect effects were estimated. The indirect impact of browsing green communities on pro-environmental behavior through environmental concern is substantial ($\beta = 0.10$, $p = 0.000$). Thus, [H3a](#) is supported. Moreover, browsing green communities had a significant indirect association with environmental concern through participation ($\beta = 0.08$, $p = 0.001$). Thus, sequential mediation of participation and environmental concern was significant ($\beta = 0.03$, $p = 0.002$), supporting [H3b](#).

Hypothesis 4 proposed that the volume of likes moderates the relationship between browsing green communities and participation in these communities. The outcomes exhibited that the volume of likes had a significant positive association with participation in green communities ($\beta = 0.38$, $p = 0.000$). In addition, the interactive term “the volume of likes*browsing green communities” was significantly related to

participation in green communities ($\beta = 0.11$, $p = 0.015$). Accordingly, [H4](#) is supported. The slope graph is presented in [Fig. 2](#). All proposed relationships were assessed based on the lower and higher volume of likes for in-depth insights. No change was observed in the significant relationship between green community browsing, participation, environmental concern, and pro-environmental behavior. However, the relationships between browsing ($\beta = 0.17$, $p = 0.069$) and participation ($\beta = 0.07$, $p = 0.532$), and environmental concern were insignificant for the lower volume of likes. For a high volume of likes, all proposed indirect relationships were significant; nevertheless, these relationships were insignificant for lower volumes of likes. Moreover, indirect effects were observed in the presence of interaction terms. The results demonstrated that browsing green communities has a significant indirect relationship with pro-environmental behavior through environmental concern ($\beta = 0.12$, $p = 0.000$). The indirect association of browsing and environmental concern through participation was also noteworthy ($\beta = 0.08$, $p = 0.001$). The sequential indirect outcome with the integration of participation and environmental concern was also substantial ($\beta = 0.04$, $p = 0.001$).

PLS-SEM focuses on the predictive significance and explains the variance of the proposed framework ([Hair et al., 2017](#)). The results suggest that the integrated constructs accounted for 35.2% of the variance in pro-environmental behavior, 35.6% in participation, and 12.9% in environmental concern. The interaction term increased the value of R^2 by 1.2% for participation in green communities (36.8%). In PLS-SEM, although the R^2 is pivotal in evaluating the proposed model, [Sarstedt, Ringle, and Hair \(2017\)](#) suggested another method to assess the predictive relevance (i.e., Q^2) by a blindfolding procedure. The values of Q^2 for endogenous constructs are above zero, i.e., 0.176, suggesting that the proposed model has good predictive relevance ([Hair et al., 2017](#)). Further, the standardized root means square residual was 0.07 for the saturated model, and NFI was 0.62. Thus, our model has a good fit since these values fall in a satisfactory category ([Hair et al., 2017](#)).

5. Discussion and implications

Green communities, like many other communities on social media, are constantly growing, and browsing social media has developed as a day-to-day activity for everyone. However, how browsing green communities affect users' pro-environmental behavior is understudied. This investigation examines the influence of browsing green communities on users' pro-environmental behavior, the underlying mechanisms, and the boundary conditions. We found that browsing green communities positively affects users' participation and environmental concerns. These findings support the literature recognizing the role of green and ethical communities in forming the ecological perception ([Cegarra-Navarro et al., 2009](#); [Hajli, 2018](#); [Hostetler et al., 2008](#)). They stated that such sustainable communities contain significant importance among the people either in the virtual or offline world and hold the potential to form their environmental concerns. It can be argued that social media's role in generating ecological discussion ([Chung et al., 2020](#)) and prominence of browsing social media (e.g., [Leong et al., 2018](#)) has been highlighted, yet, findings relevant to green communities offer in-depth and lucid insights. Further, the results indicate that participation in green communities exerts a strong positive effect on environmental concerns. Participation is an important pathway linking green community browsing and environmental concerns. Broadly, the results are similar to [Cegarra-Navarro et al. \(2009\)](#), suggesting that individuals' participation is critical in promoting environmentally friendly behavior while living in green communities since it influences their environmental knowledge and concerns. These results also agree with social media literature; for instance, [Lee et al. \(2020\)](#) and [Zafar, Shen, Ashfaq, and Shahzad \(2021\)](#) highlight the essentiality of social networking platforms and social ties in altering netizens' environmental responsibility and participation in recycling campaigns. Nevertheless, our findings unveil noteworthy insinuations of social media green

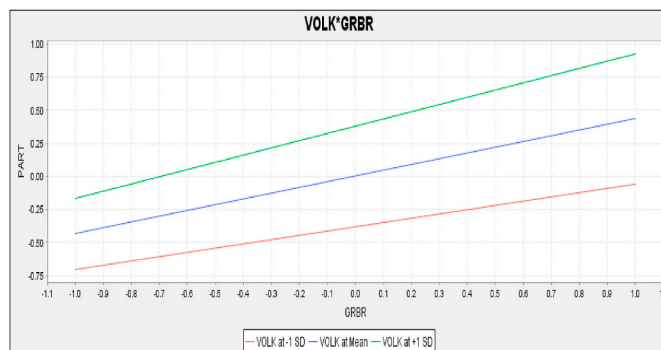


Fig. 2. Moderation of the volume of likes. Note: VOLK = volume of likes, GRBR = Browsing of green communities, PART = participation in green communities.

communities.

Furthermore, the results demonstrate that environmental concerns exert a positive influence on pro-environmental behavior and that environmental concerns mediate the relationship between green community browsing, participation, and pro-environmental behavior. These findings endorse the extant literature on individuals' environmental concerns acting as a significant predictor of eco-friendly attitudes, intentions, and behavior (e.g., Alzubaidi et al., 2020; Cerri et al., 2018; Lee et al., 2014; Yadav & Pathak, 2016). Our results on the sequential mediation of participation and environmental concern in encouraging pro-environmental behavior stemming from browsing green communities are novel. Further, our study also identifies a significant positive interaction role of the volume of likes in the connection between browsing and participation in green communities. These results agree with Zafar et al. (2019), suggesting that social media users observe these numeric features, which may influence their cognitive and affective behaviors in the virtual environment.

5.1. Theoretical implications

This study contributes in numerous ways. First, a research model is developed in the context of green communities to capture the behavioral consequences for their members. Social learning theory underpins the proposed framework, presents several new correlates, and extends social learning theory to the green community framework. This novel study reveals that social learning plays a vital part in identifying pro-environmental actions as an effect of social networking sites. Second, this research contributes to the social media literature relevant to green communities, which emerge as a predictor of their members' pro-environmental behavior. The extant literature provides empirical evidence regarding mass media use in promoting environmentally friendly behavior (Junsheng, Akhtar, Masud, Rana, & Banna, 2019). Nevertheless, pro-environmental behavior in the context of browsing green communities is underexplored, so the investigation of the underlying mechanisms of browsing green communities is a crucial step in understanding the social development of how social networking platforms affect users' behavior.

Third, the probable repercussions of browsing green communities on environmental concern and participation have not been investigated. Our results can be related to those highlighting the ascendancy of social networking platforms' usage in promoting environmental education and recycling actions (Chung et al., 2020; Sujata, Khor, Ramayah, & Teoh, 2019). The virtual environment of social media, including different online communities, has emerged as a powerful tool to shape their members' behavior (Brown, 2015; Zafar et al., 2020, 2021). Researchers have emphasized the prominence of participation in building consumers' ethical perceptions (Nadeem et al., 2019) and buying behavior (Leong et al., 2018). They suggested that participation in social media affects users' social behavior, but there has been little exploration of the underlying mediating mechanisms. This study demonstrates that social media participation affects users' social behavior by affecting their social attitudes, e.g., environmental concerns. Moreover, posts with higher numbers of likes draw more attention from community members, effectively enhance their ecological concerns, and ultimately encourage pro-environmental actions. The direct impact of observational learning has been highlighted in several studies (Chen et al., 2016; Zhang et al., 2018). This research emphasizes the role of interaction in perception, contributing to the business ethics literature by developing a fundamental cognitive information mechanism for promoting pro-environmental behavior on social media. It also contributes to the perspective that the overall impact of netizens' presence on social media must be recognized distinctively compared to the specific media, such as green communities.

5.2. Practical implications

This research offers novel insights to articulate effective social media policies relevant to green communities and pro-environmental behavior. First, the significant impact of browsing green communities indicates that this exploratory behavior needs to be encouraged; for instance, social media marketing of green communities and their integration into other netizens' digital aspects, such as environmental games, will promote green communities and eco-friendly concerns. Further, green community administrators should be engaged as environmental "evangelists" by organizations, governments, and NGOs. These people can play an influential role, and concerned stakeholders, including brands, may have a dedicated network of effective pro-environmental advocates. Second, our findings relevant to the volume of likes indicate that these cues promote participation and influence participation. Accordingly, policymakers may sponsor these communities and take the paid option for certain posts, as social media provides facilities that help attract more likes. In turn, this will effectively instigate environmental behavior.

Third, our findings indicate that members of green communities are inclined to interact and support each other to encourage environmental concerns, leading to pro-environmental behavior. Social media platforms (e.g., Facebook) categorize users considering the importance of their participation and provide the badge of "Top Fans" to the most active participants, which is an intensive process. These "Top Fans" actively participate in different activities such as reacting, liking, sharing, or commenting on the content encountered. These fans might be supporting environmental campaigns; however, on the other side, many active participants can have a pessimistic perspective regarding ecological behavior. They both have distinguished roles but need to be identified and addressed differently. For instance, "Top Fans" with an optimistic environmental approach can be a worthy supporters to endorse eco-friendly behavior. Accordingly, they need to be tracked and promoted by concerned stakeholders to encourage participation and change the perception regarding peers' environmental behavior, including their interpersonal network of friends and family. In contrast, abrogating perspective attracts a lot of attention nevertheless, high environmental skepticism might be entrenched with a negative experience or lack of understanding about sustainable development goals. Those pessimistic fans should not be disregarded; their concerns must be addressed. Like, these participants should be engaged not only personally by the stakeholders; instead, a focus group can be established to address their issues effectively, surpassing their expectations. This personal immersion will positively shape environmental concerns and behaviors. Following the potential of green communities, interactive social media movements utilizing the hashtag feature could be launched to promote participation and ecological concerns that will ultimately assist in meeting sustainable development goals.

5.3. Limitations and future research directions

This study has a few limitations that need to be addressed. Many green community members might have adverse perspectives regarding environmental approaches and can be involved in negative participation, which is not explored in this study. Such encountered circumstances may alter pro-environmental behavior; therefore, the extension of the proposed framework might offer valuable insights. Many people are not well aware of ecological problems, so their incorporation as a control group through an experimental approach can assist in designing efficient pro-environmental strategies. Furthermore, a substantial number of netizens do not follow green communities; thus, investigating followers and non-followers will allow better recognition of the overall impact of social networking platforms in the milieu of pro-environmental behavior. Other numeric cues (e.g., the volume of shares and comments) also exist in the platform of social media green communities, which can be integrated into future studies for more

insights. Lastly, the data originated from a single country, and results may vary due to distinctive cultural values, beliefs, and diverse usage patterns of social media in different populations. An in-depth comparative analysis will offer better pro-environmental implications and a more generalizable strategy.

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On behalf of all authors, the corresponding author state that the authors have no competing interests to declare that are relevant to the

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Credit author statement

Jie Shen: Supervision, Project administration, funding acquisition, Revising, and writing -original draft. Hanyuan Liang: Conceptualization, Methodology, Software, Writing - original draft. Abaid Ullah Zafar: Formal analysis, Investigation, Revising. Mohsin Shahzad: Resources, Validation. Umair Akram: Writing - review & editing. Muhammad Ashfaq: Writing – Revising & Proof reading.

Data availability

Data will be made available on request.

Appendix



Source: Environmental Change Makers; Save the Environment

Table 1
Respondents' attributes (n = 585)

Attributes	Distribution	Frequency	Percent
Gender	Male	323	55.2
	Female	262	44.8
Age	Under 19 years	79	13.5
	19–36 years	336	57.4
	37–54 years	170	29.1
Qualification	High school or below	16	2.7
	Two years college	58	9.9
	Bachelors	212	36.2
	Masters	206	35.2
Occupation	PhD	93	15.9
	Student	299	51.1
	Employee	210	35.9

(continued on next page)

Table 1 (continued)

Attributes	Distribution	Frequency	Percent
Social media usage	Self employed	46	7.9
	Unemployed	30	5.1
	Everyday	388	66.3
	Several times a month	69	11.8
	Once a week	21	3.6
Income (USD)	Several times a week	107	18.3
	Less than or equal to 200	62	10.6
	201–400	148	25.3
	401–600	123	21.0
	601–800	101	17.3
	801–1000	94	16.1
No of following communities	More than 1000	57	9.7
	Less than 5 communities	99	16.9
	5–10 communities	280	47.9
	More than 10 communities	206	35.2

Table 2

Constructs' reliability and validity

Constructs	Composite reliability	Average variance extracted	Factor loadings	Mean	Standard deviation
Browsing of green communities (GRBR)	0.91	0.77	–	–	–
GRBR1			0.88	5.23	1.11
GRBR2			0.85	5.21	1.08
GRBR3			0.89	5.24	1.18
Participation in green communities (PART)	0.88	0.54	–	–	–
Actionable participation (ACPC)	0.89	0.80	–	–	–
ACPC1			0.88	5.22	1.15
ACPC2			0.91	5.39	1.16
Attitudinal participation (ATPC)	0.87	0.78	–	–	–
ATPC1			0.88	5.17	1.09
ATPC2			0.87	5.50	1.08
Informational participation (INPC)	0.92	0.85	–	–	–
INPC1			0.92	5.12	1.13
INPC2			0.91	5.03	1.07
Environmental concern (ENVC)	0.88	0.64	–	–	–
ENVC1			0.80	5.35	1.26
ENVC2			0.79	5.61	1.06
ENVC3			0.86	5.03	1.26
ENVC4			0.76	4.99	1.08
Pro-environmental behavior (PREV)	0.89	0.52	–	–	–
PREV1			0.75	5.19	1.26
PREV2			0.71	5.37	1.18
PREV3			0.71	4.96	1.22
PREV4			0.72	4.95	1.24
PREV5			0.69	5.51	1.12
PREV6			0.74	5.26	1.20
PREV7			0.73	5.22	1.15
Volume of likes (VOLK)	0.93	0.81	–	–	–
VOLK1			0.87	5.71	0.90
VOLK2			0.93	5.72	0.88
VOLK3			0.90	5.77	0.94

Table 3

Fornell-Larcker criterion

	ACPC	ATPC	ENVC	GRBR	INPC	PREV	VOLK
ACPC	0.89						
ATPC	0.53	0.88					
ENVC	0.27	0.26	0.80				
GRBR	0.33	0.42	0.32	0.88			
INPC	0.45	0.54	0.19	0.40	0.92		
PREV	0.32	0.30	0.51	0.45	0.33	0.72	
VOLK	0.37	0.38	0.13	0.11	0.23	0.287	0.90

Note: Bold values are the square root of relevant AVE. ACPC-actionable participation; ATPC-attitudinal participation; ENVC-environmental concern; GRBR-browsing of green communities; INPC-informational participation; PREV-pro-environmental behavior; VOLK-volume of likes.

Table 4
Indicator correlations

	ENVC	GRBR	PART	PREV	VOLK
ENVC	1	0.32	0.29	0.51	0.13
GRBR	0.32	1	0.47	0.45	0.11
PART	0.29	0.47	1	0.38	0.42
PREV	0.51	0.45	0.38	1	0.26
VOLK	0.13	0.11	0.42	0.26	1

ENVC-environmental concern; GRBR-browsing of green communities; PART-participation; PREV-pro-environmental behavior; VOLK-volume of likes.

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